

Ion Exchange in Sodium Borosilicate Glasses

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The kinetics of K^+ - Na^+ exchange in sodium borosilicate glasses containing 20 mol% Na_2O were studied as a function of glass composition and exchange time and temperature. The distribution of K in the glass after exchange in molten KNO_3 was determined using an electron microprobe. In those specimens which were not chemically attacked by the molten salt, the K profiles were consistent with Fickian behavior. The calculated interdiffusion coefficients varied with local composition in each specimen. This composition dependence and its variation with temperature could be fitted approximately to a mixed-alkali model using the Nernst-Planck relation for diffusive fluxes. However, some deviation from this behavior, presumably as a result of the generation of microscopic stresses around an exchange site, was observed. The dependence of the apparent activation energy, $\bar{E}$, on the B/Si ratio was analogous to that of the activation energy for ionic conduction in these glasses: $\bar{E}$ decreased with increasing fraction of nonbridging oxygen. This result disagrees with the presumed behavior in sodium aluminosilicate glasses.

I. Introduction

STRENGTHENING of glass by ion exchange is generally conducted by immersing it in a molten salt bath at a temperature lower than the glass transition temperature, T_g . The salt is usually that of an alkali whose ionic size is larger than that of the alkali in the glass. Two processes accompany the exchange, i.e. (1) generation of stresses within the glass because of the size difference of the ions and (2) relaxation of these stresses through glass-network dilatation to accommodate the size difference. A model for the latter process is provided by Hale,¹ who followed Eshelby's analysis.² Spoor and Burggraaf³ and Garfinkel and King⁴ attempted to describe stress relaxation mathematically. The generation of stresses by ion exchange is diffusion-dependent. Diffusion kinetics during ion exchange in glass have been studied by several workers⁵⁻⁹ as a function of such experimental variables as glass composition, salt-bath composition, and exchange time and temperature. The elucidation of the basic mechanism of ion exchange has been one of the primary aims of most of these studies.

The present work focuses on the kinetics of ion exchange as a function of the experimental variables just mentioned. At the same time, it also examines the influence of nonbridging oxygen in glass on the kinetics of ion exchange since it has been suggested that, in alkali aluminosilicates, the disappearance of nonbridging oxygen at a molar ratio $Al/Na=1$ may explain¹⁰ the maximum in electrical conductivity,¹⁰⁻¹² the self-diffusion coefficient of the alkali,¹¹ and the interdiffusion coefficient for ion exchange.⁷ Evidence based on measurements of physical properties,¹³ in particular internal friction,^{13,14} has been put forward to support the notion that nonbridging oxygen in the aluminosilicates disappears at $Al/Na=1$. However, the validity of some conclusions based on internal-friction measurements has been questioned.^{15,16} A study of the variation of nonbridging oxygen with composition in the Na_2O - B_2O_3 - SiO_2 system is likely to be more nearly definitive than such a study in the aluminosilicates because 3- and 4-coordinated boron can be determined by NMR.¹⁷

According to Milberg *et al.*,¹⁷ at a given concentration of alkali, the fraction of nonbridging oxygen in the Na_2O - B_2O_3 - SiO_2 system depends on the B/Si ratio. On the other hand, in alkali-lime silicates, the fraction of alkali fixes the minimum value for the fraction of nonbridging oxygen. The fact that nonbridging oxygen in the borosilicates can be varied similarly to its variation in the aluminosilicates was the reason for selecting the Na_2O - B_2O_3 - SiO_2 system for the present study.

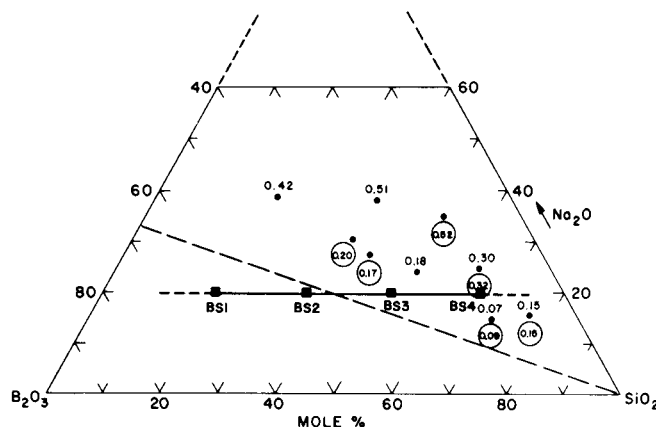


Fig. 1. Variation of mole fraction of nonbridging oxygen with composition in Na_2O - B_2O_3 - SiO_2 system (Ref. 17) and compositions of glasses studied.

The kinetics of Na^+ - K^+ exchange in the Na_2O - B_2O_3 - SiO_2 system were studied as a function of exchange temperature (340° to $500^\circ C$), exchange time (18 to 3600 h), and glass composition (fixed Na content and varying B/Si ratio). Except for one experiment where a mixture of KNO_3 and $NaNO_3$ was used, glass specimens were exchanged in a bath of molten KNO_3 . Results of a few experiments where glasses of the Na_2O - K_2O - B_2O_3 - SiO_2 system were exchanged in molten KNO_3 are also reported.

II. Experimental Procedure

The sodium borosilicate glass compositions chosen for study are marked on Fig. 1, which shows the variation of nonbridging oxygen fraction in the Na_2O - B_2O_3 - SiO_2 system according to Milberg *et al.*¹⁷ The compositions lie on a 20 mol% Na_2O line away from the region of phase separation.¹⁸ Thus, the glasses contain a constant number of mobile species (Na^+) with a varying fraction of nonbridging oxygen.

Batches corresponding to 600 g of glass each were prepared from reagent-grade chemicals. A small amount of colorant was added to each to identify the end product. Melting was done in a Pt20Rh crucible inside an electric resistance furnace at 1100° to $1550^\circ C$ (the higher temperatures were necessary for the higher SiO_2 compositions). Melts were stirred occasionally with a fused- SiO_2 rod. After sufficient time for fining was allowed, the crucibles were taken out of the furnace and cooled slowly. Glass blocks removed from the crucibles by tapping were then annealed. Specimens 8 by 8 by 4 mm were cut from these blocks. The large faces were ground flat and polished using 3- μm diamond abrasive.

Table I shows the glass compositions, T_g obtained by DTA, density, and fraction of nonbridging oxygen (measured by NMR). Electron micrographs and X-ray powder patterns of each glass showed no signs of phase separation or devitrification. Electron-microprobe analysis for Na and Si in pieces randomly picked from the bulk glass assured homogeneity within each melt.

The apparatus for ion exchange (Fig. 2) consisted of a temperature-controlled ($\pm 1.5^\circ C$) vertical tube furnace in which a

Presented at the 74th Annual Meeting, The American Ceramic Society, Washington, DC, May 10, 1972 (Glass Division, No. 51-G-72). Received March 28, 1973; revised copy received October 24, 1973.

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Table I. Properties of Glasses Studied

Glass	Nominal mol%			Wt%			T_g	Density (g/cm ³)	Fraction nonbridging oxygen
	Na ₂ O	B ₂ O ₃	SiO ₂	Na ₂ O	B ₂ O ₃	SiO ₂			
BS1	20	60	20	18.75 (18.5)	63.1 (63.9)	18.15 (17.8)	508°C	2.25	0
BS2	20	45	35	19.15 (18.8)	48.4 (48.6)	32.45 (32.6)	545°C	2.37	0
BS3	20	30	50	19.6 (18.8)	33.0 (35.3)	47.4 (46.1)	577°C	2.35	0.19
BS4	20	15	65	20.05 (19.7)	16.85 (17.0)	63.1 (63.3)	600°C	2.49	0.32

*Numbers in parentheses are values obtained by analysis.

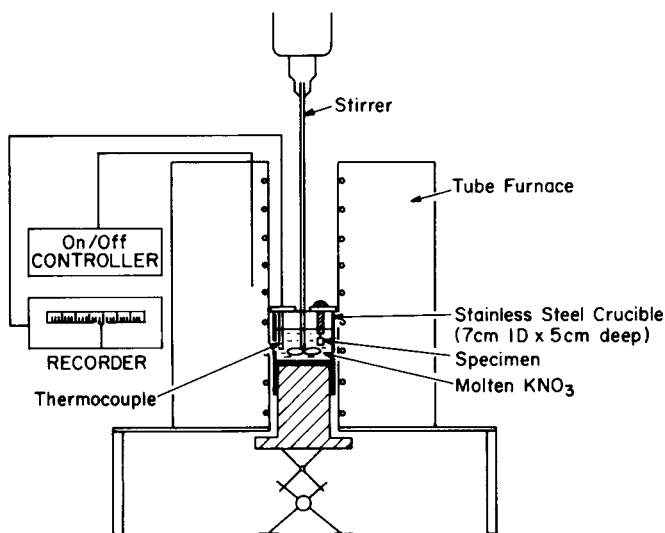


Fig. 2. Ion-exchange apparatus.

stainless-steel crucible containing the molten salt was placed. The salt was continuously stirred and its temperature recorded. Four specimens could be ion exchanged at once.

After ion exchange was conducted for the desired period, the specimens were removed from the furnace, cooled, washed to remove the adhering salt, and sliced along the direction of diffusion. The ion-exchanged faces of a slice from the middle of the specimen (avoiding end effects) were covered with Pt or Al foil. The slice was then mounted in a plastic* medium with one of the faces containing the diffusion direction exposed. The mount was ground flat, polished, coated with C film ≈ 300 Å thick, and analyzed for the distribution of K in the diffusion zone with an electron probe.[†]

For analysis, the electron beam was focused to a line ≈ 80 μ m long oriented perpendicular to the direction of diffusion while the specimen was stepped at intervals of 1 or 2 μ m in the direction of diffusion. Characteristic X-ray intensities for K were collected at each step for a preset probe current. The counting time corresponding to this current was ≈ 8 s.

Counts were converted to wt% oxide by using a set of glass standards with a varying ratio of K₂O to Na₂O; the ones based on the matrix of glass BS3 are listed in Table II. Composition C55 is the potassium equivalent of the BS3 mole composition. If all the Na⁺ in BS3 were replaced by K⁺, the wt% alkali in the glass would increase from 19.6 (BS3) to 27.1 (C55).

The matrix effect on the electron-probe analysis for K was negligible. However, whereas the relation between counts and analyzed wt% K₂O using 2- μ m steps was a straight line, departure from linearity at higher concentrations of K as a result of increased ionic migration¹⁹ occurred when 1- μ m steps were used.[‡] Since the onset of ionic migration may disturb the K distribution in glass, analyses

using 1- μ m steps were avoided as much as possible and confined to the surface region, where spatial resolution to determine steep concentration gradients was important.

III. Calculations

When only one rate-controlling diffusion coefficient operates within a system, the diffusion equation is

$$\frac{\partial C_{K_2O}}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C_{K_2O}}{\partial x} \right) \quad (1)$$

where C is the concentration (g/cm³ herein), x the distance (cm) from the surface into the glass, t the time (s), and D the diffusion coefficient (cm²/s).

When D is constant, the solution²⁰ of Eq. (1) with initial and boundary conditions of a unidirectional semiinfinite couple (appropriate for the experiment) is

$$C_{K_2O,t} = C_0 \operatorname{erfc} (x/2\sqrt{Dt}) \quad (2)$$

where C_0 = surface concentration of K₂O. When D depends on C , the concentration distribution is no longer described by Eq. (2). In such cases, the usual method of determining the concentration-dependent D is to apply the Boltzmann-Matano technique²¹ through which $D(C)$ is determined as

$$D(C) = - (1/2t) \left\{ \left(\int_0^x x dC \right) / (dC/dx) \right\} \quad (3)$$

The diffusion coefficient, $D(C)$, thus obtained is essentially a local diffusion coefficient.

Since the width of the electron-beam-excited volume of glass was ≈ 2 to 3 μ m, the distribution of K in ion-exchanged samples, as measured by the microprobe, required deconvolution to obtain the true distribution.²² This deconvolution was achieved using Mencik's modified Fourier technique²³ with a provision for a measured beam function which varied as a result of density variations in the zone analyzed.²⁴ The extremes in this varying beam function were obtained by measuring the K₂O distribution across composites made by pressing a flat, polished face of a potassium borosilicate glass (C55) against (1) a flat, polished face of a sodium borosilicate glass (BS3) and (2) Pt or Al foil. The deconvolution procedure also established the location of the glass surface on the plots of measured wt% K₂O vs distance in the diffusion direction to within 0.5 μ m.

Following deconvolution, conversion from wt% K₂O to concentration units, i.e. g K₂O/cm³, was achieved using a density correction based on Hale's analysis.¹ For example, if no network dilatation occurred in glass BS3 on ion exchange, the fully exchanged surface layer of the glass would increase 10% in mass density. Hale

Table II. Calibration Glasses for Microprobe

Glass	Component (mol%)			
	Na ₂ O	K ₂ O	B ₂ O ₃	SiO ₂
BS3	20	0	30	50
C1	16	4	30	50
C2	12	8	30	50
C3	8	12	30	50
C4	4	16	30	50
C55	0	20	30	50

NOTE: The wt% composition (analyzed) of C55 is 27.1% K₂O, 30.0% B₂O₃, 42.9% SiO₂.

*Bakelite, Union Carbide Corp., New York, NY.

†Model EMX, Applied Research Laboratories, Sunland, CA.

‡Microanalysis for Na distribution was not performed because the calibration curve for Na was poor.

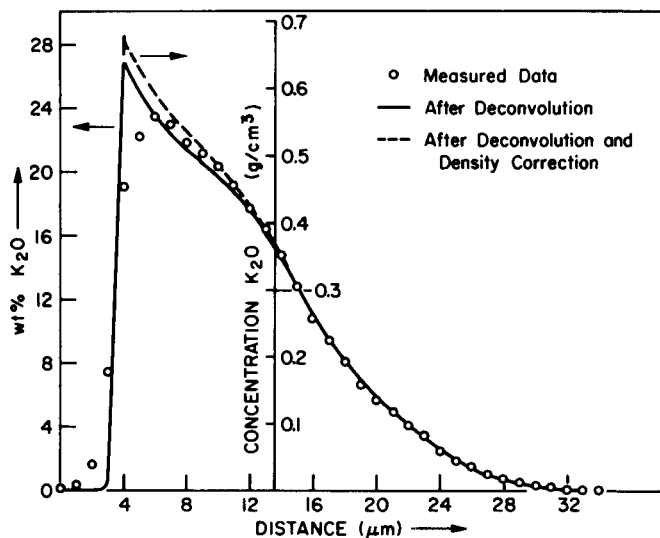


Fig. 3. Distribution of K_2O in glass BS3 after ion exchange at $473^\circ C$ for 18 h.

noted that a network dilatation of $2.1 \text{ cm}^3/\text{mol}$ of exchanged Na corresponded well with the observed value of surface compression in ion-exchanged specimens. The same value for the dilatation corresponded to the compressive stresses observed in the present specimens; hence, it could be shown that the fully exchanged surface of glass BS3 densified by 6.8%. Assumption of a linear relation between the fraction exchanged and densification achieved then gave the required density correction.

After the density correction was applied, the profiles of K_2O concentration vs diffusion distance departed, in general, from Eq. (2). The diffusion coefficient, therefore, was calculated at each K_2O concentration using Eq. (3). An effective diffusion coefficient, $\bar{D}$, called the "mass average diffusion coefficient" and defined as

$$\bar{D} = (\pi/t)(M/2C_0^2) \quad (4)$$

where M = total mass of K_2O absorbed/unit area of the exchanging surface, was also calculated for all exchange experiments.

IV. Results

In Fig. 3, the measured wt% K_2O is plotted vs distance from an arbitrary origin in a typical specimen. The standard deviation associated with each of the points is less than the radius of the circles. The results after deconvolution and after deconvolution and density correction are also shown. The solid line indicates that the value of

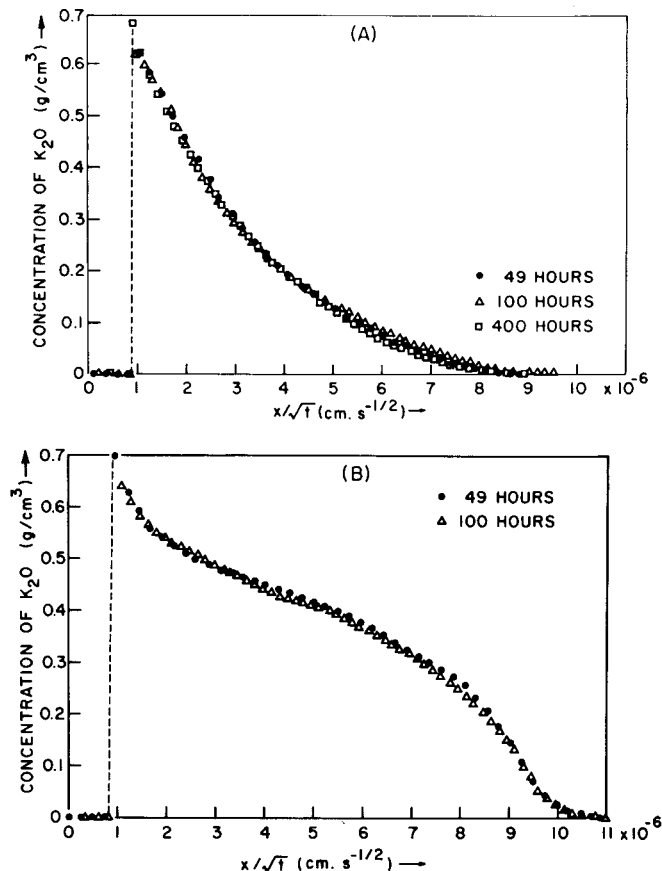


Fig. 4. Concentration of K_2O vs $x/\sqrt{t}$ for (A) glass BS4 exchanged at $400^\circ C$ and (B) glass BS1 exchanged at $450^\circ C$.

C_0 in this case was 27.0 wt%, which corresponded to the value in the fully K-exchanged glass (C55). In most cases, the calculated value of C_0 after deconvolution was within 1 wt% of the expected value.

To test the Fickian nature of the diffusion coefficient, the concentration of K_2O was plotted vs $x/\sqrt{t}$, where a particular glass was ion exchanged for several values of t at a given temperature. Typical results are shown in Fig. 4.

Figure 5 shows the values of $D(C)$ in the 4 glasses calculated by the Boltzmann-Matano method for exchanges at 340° to $500^\circ C$ for up to 3600 h. The ordinate is $\log D(C)$ and the abscissa the fractional exchange, γ , where $\gamma = C_{K_2O}/(\text{fully K-exchanged value})$

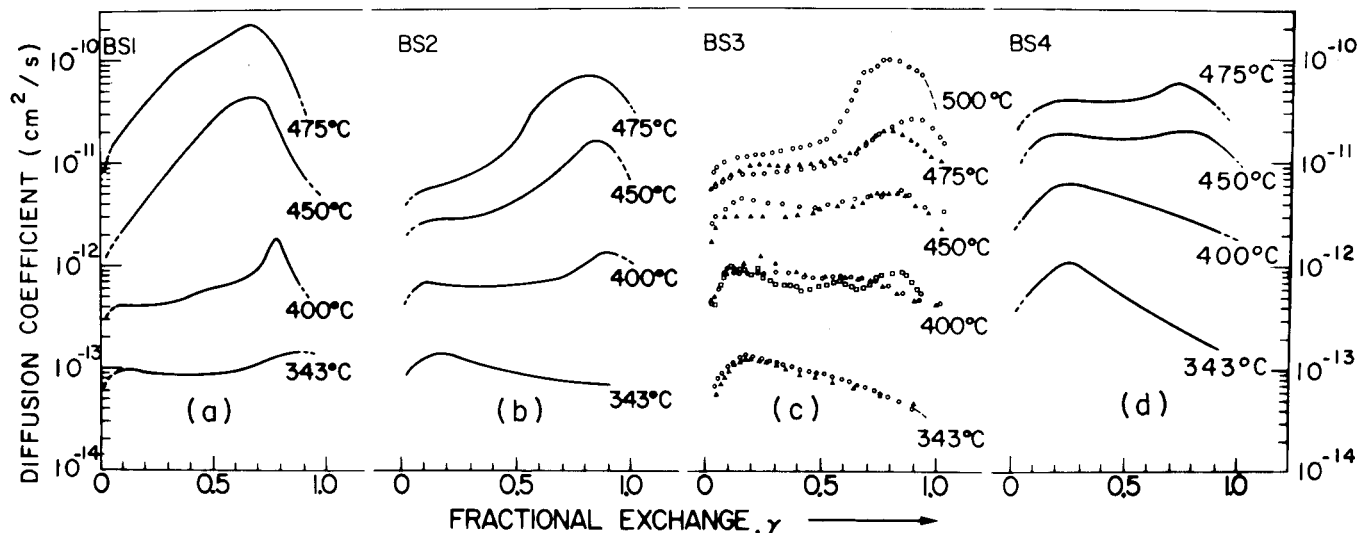


Fig. 5. Variation of D with local composition, temperature, and time for each glass. For glass BS3, ($500^\circ C$) $\circ$ = 49 h; ($475^\circ C$) $\circ$ = 49 h, $\triangle$ = 18 h; ($450^\circ C$) $\circ$ = 49 h, $\triangle$ = 102 h; ($400^\circ C$) $\circ$ = 100 h, $\triangle$ = 91 h, $\square$ = 400 h; ($343^\circ C$) $\triangle$ = 964 h, $\circ$ = 3600 h.

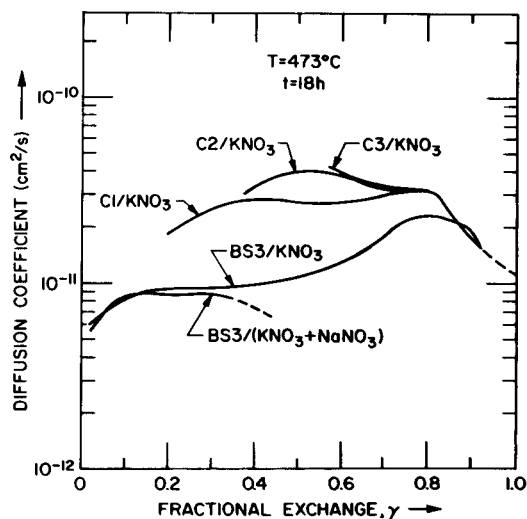


Fig. 6. Variation of D with local composition in mixed-alkali glasses ion-exchanged with KNO_3 and glass BS3 ion-exchanged with $(\text{KNO}_3 + \text{NaNO}_3)$.

of $C_{\text{K}_2\text{O}}$ in the glass). For clarity, detailed data are shown for glass BS3 only. Because of the detection limit of the microprobe and the inadequacy of deconvolution near the surface, the calculated values of $D(C)$ are meaningful only between $\gamma = 0.05$ and 0.9 . It is estimated that the individual values of $D(C)$ may be in error by as much as 50%. No estimate of errors for the pooled values of $D(C)$ on a particular curve was attempted.

Figure 6 is a plot similar to Fig. 5 in which results are shown for those experiments where (1) mixed-alkali glasses based on the matrix of BS3 (Table II) were exchanged at 473°C in molten KNO_3 and (2) glass BS3 was exchanged at 473°C in a mixture of 80 mol% KNO_3 and 20 mol% NaNO_3 .

Chemical attack, indicated by formation of a water-insoluble white deposit on the surface of the glass, occurred in some specimens. The extent of attack increased with temperature and with B/Si ratio. The results in Figs. 5 and 6 do not include those for samples which were attacked chemically to a significant extent, i.e. those where the white deposit was visible to the naked eye.

Figure 7 is a semilogarithmic plot of the mass average diffusion coefficient, $\bar{D}$, vs the reciprocal of absolute temperature. Except for the 343°C value of glass BS1, all data fall on straight lines of the Arrhenius type. The values of the preexponential factor, $\bar{D}_0$, and the apparent activation energy, $\bar{E}$, which fit these lines are given in Table III. (The calculation of $\bar{E}$ for BS1 ignored the measured $\bar{D}$ at 343°C .)

V. Discussion

The concentration distributions of K_2O in a given glass corresponding to various exchange times at a given temperature merge when the abscissa is $x/\sqrt{t}$, regardless of the shapes of the distributions (Fig. 4). This behavior shows that, although the diffusion coefficient is in general a function of local concentration, it is independent of both the exchange time and the concentration gradients, i.e. the ion exchange is a Fickian phenomenon.

In a separate experiment with grains of glass BS4 (to expose a large surface area for exchange), no significant traces of B and Si could be detected in the molten KNO_3 after ion exchange. This result indicates that ion exchange is adequately described by a one-to-one exchange between Na^+ and K^+ ions. Since there is only one independent concentration gradient* during ion exchange in glass, Eq. (1) with only one rate-controlling coefficient describes the process. As such, the local diffusion coefficient used in Eq. (1) is identical to the interdiffusion coefficient for ion exchange.

It is clear from Fig. 5 that the interdiffusion coefficient, $D(C)$, in a given sample varied significantly with γ . An activation energy,

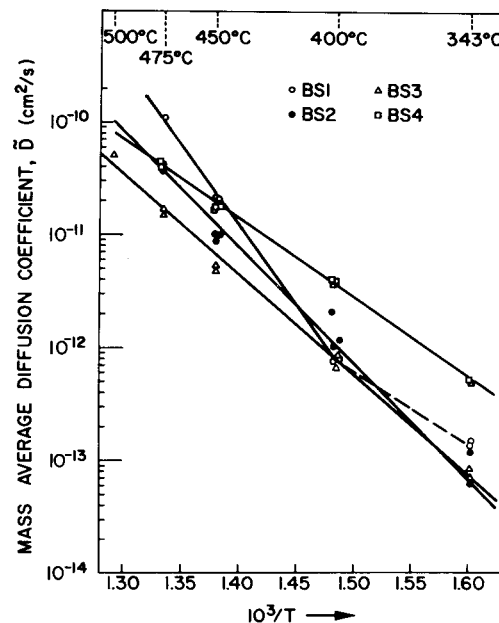


Fig. 7. Variation of $\bar{D}$ with $1/T$.

Table III. Preexponential Factors and Apparent Activation Energies in Arrhenius Equation $\bar{D} = \bar{D}_0 \exp(-\bar{E}/RT)$

Glass	$\bar{D}_0$ (cm^2/s)	$\bar{E}$ (kcal/mol)
BS1	3.9×10^{10}	71.0
BS2	2.2×10^3	47.5
BS3	14.3	41.1
BS4	0.11	32.5

$E(C)$, can be calculated to describe the temperature dependence of $D(C)$ in a given glass. Values of $E(C)$ for glass BS3, for example, are 26, 35, 43, and 55 kcal/mol at $\gamma = 0.2, 0.4, 0.6$, and 0.8 , respectively, indicating that $E(C)$ as well as $D(C)$ varies with γ .

Variation of $D(C)$ in the ion-exchanged region of a glass was encountered by Doremus.⁵ Assuming that the ions exchanged with each other to minimize the local buildup of electric charge, he calculated $D(C)$ using the Nernst-Planck equation⁵:

$$D(C) = (D_{\text{Na}} D_{\text{K}}) / (D_{\text{Na}} N_{\text{Na}} + D_{\text{K}} N_{\text{K}}) \quad (5)$$

where D_i = the self-diffusion coefficient of ion i in the base glass and N_i = (moles of i) / (total moles of alkali in glass). Good agreement between measured and calculated $D(C)$ values was found in Doremus' experiments. Lacharme²⁶ modified Eq. (5) by defining D_i as the self-diffusion coefficient of i not in the base glass but in a melted glass composition[†] corresponding to γ . Lacharme compared the calculated $D(C)$ using self-diffusion data (and Eq. (5)) and the measured $D(C)$ from ion-exchange experiments directly. The comparison, made at only one temperature, revealed that, away from the surface region, the agreement between the two was excellent. Hayami and Terai²⁷ also reported general agreement of the measured $D(C)$ with those calculated from self-diffusion data. The results were reported, again, for only one temperature.

Extending the applicability of Lacharme's argument²⁶ to all temperatures leads to the hypothesis that, as far as diffusion behavior is concerned, ion-exchanged glass in the diffusion zone approximates a composite of stacked layers of mixed-alkali glasses with a gradually varying ratio of the alkalis. Since self-diffusion data in the $\text{Na}_2\text{O}-\text{K}_2\text{O}-\text{B}_2\text{O}_3-\text{SiO}_2$ system are lacking, this hypothesis (the "mixed-alkali model") cannot be tested directly.

*It is assumed that deviations from local stoichiometry are negligible during interdiffusion in glass (Ref. 25).

† N_K in a melted glass = γ in an ion-exchanged glass.

However, self-diffusion coefficients in a series of mixed-alkali compositions have been measured in several other glass systems.^{26,28-31} A calculation of $D(C)$ in these systems at various temperatures and for varying alkali ratios clearly shows that both $D(C)$ and $E(C)$ vary with alkali ratio in a manner roughly similar to that observed in the present study.

However, the hypothesis of the mixed-alkali model cannot be altogether true. Lacharme²⁶ observed a discrepancy between the measured and calculated $D(C)$ values in the surface region which he ascribed to a "changed structure" at the surface. Hayami and Terai²⁷ ascribed their observed differences to a negative departure from Raoult's law. Figure 6 shows that $D(C)$ is not a unique function of γ but depends on the alkali ratio in the base glass as well as that in the salt bath. It is difficult to reconcile these observations with complex forms^{32,33} of the Nernst-Planck equation involving chemical-activity effects alone.* It may be that the Nernst-Planck equation should be modified to account for stress effects as well. However, it can be argued that the role of macroscopic stresses (compression in the outer layers and tension in the interior) must be small because identical diffusion coefficients are obtained (Fig. 4) in specimens with widely varying exchange times (and hence varying levels of achieved stresses, as observed in a few selected specimens). Furthermore, the macroscopic stresses cannot be responsible for the concentration dependence of D itself, contrary to the conclusions of Chakravorty *et al.*³⁵ It is clear from the present results for glass BS1 ($T_g = 508^\circ\text{C}$), from Lacharme's results,²⁶ and from the measurements of Hayami and Terai²⁷ (all conducted in the vicinity of T_g) that concentration dependence of D may be expected even at temperatures where the glass body cannot sustain a significant level of macroscopic stresses.

It is likely that the necessary modifications in the mixed-alkali model require consideration of the microscopic stresses generated around an exchange site. Figure 6 indicates that exchange between a foreign ion, B, and a host ion, A, is easier when the host structure corresponds more to B-containing glass than to A-containing glass. The ease may result from diminished production of microscopic stresses around an exchange site. However, it is not at all clear whether an increased ability to accommodate an exchanging ion by network dilatation or an initial reduction in the difference of the sizes of the exchanging ion and the average host site in the network causes the diminished production of stresses.

Values of the apparent activation energy, $\bar{E}$, appear rather high compared to the activation energies involved in the motion of alkali ions in either self-diffusion or electrical conduction. However, $\bar{D}$ has not been defined on an atomistic basis; it involves a complex averaging of the local values of the interdiffusion coefficient. Nevertheless, high values of $\bar{E}$ have been reported by others.³⁶ Also, the pattern of variation in $\bar{E}$ with B/Si ratio is similar to that observed by Otto³⁷ for the activation energy for electrical conduction in the $\text{Na}_2\text{O}-\text{B}_2\text{O}_3-\text{SiO}_2$ system. No positive correlation between the fraction of nonbridging oxygen and the activation energy similar to that believed to exist in the aluminosilicates is seen. Although some caution should be used in interpreting $\bar{E}$ because of its nonatomistic nature, it is tempting to conclude that the relation between nonbridging oxygen and the activation energy for alkali motion is not one of cause-and-effect.

It may be noted from Fig. 5 that the temperature at which the diffusion coefficient is apparently independent of γ increases with Si/B ratio. Along the same direction in composition space, the glass transition temperature, the resistance to chemical attack, and the fraction of nonbridging oxygen also increase. The relation, if any, among these observations, however, is not clear.

VI. Conclusions

Ion exchange in the $\text{Na}_2\text{O}-\text{B}_2\text{O}_3-\text{SiO}_2$ system over the range of temperature, time, and compositions studied is a Fickian phenom-

non. The diffusion coefficients which describe the rate of ion exchange are in general concentration-dependent. The concentration dependence of these coefficients is similar to that predicted by the mixed-alkali model proposed by Doremus⁵ and by Lacharme,²⁶ although this model must be modified to account for differences between the network of an ion-exchange-produced composition and that of a corresponding batch-melted composition. The relation between the fraction of nonbridging oxygen in the compositions studied and the apparent activation energy for the overall kinetics of ion exchange is not one of cause-and-effect.

Acknowledgments: The writers are indebted to R. Gardon for critical evaluation and to E. Morgan for assistance with the experimental work.

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*Such equations have been used successfully to explain the results of ion exchange in zeolite systems (Ref. 34).